

EDA Financial Discussions - Analysis Report

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Executive Summary

This report summarizes the exploratory data analysis conducted to identify suitable datasets for predicting engagement and sentiment surges in stock-related social media discussions.

Key Findings

- **Datasets discovered:** 0 (0 complete with engagement + sentiment fields)
- **API platforms assessed:** 2
- **Quality reports generated:** 1 (1 suitable)
- **Surge definitions evaluated:** 9 (6 viable with $\geq 2\%$ positive class)

Dataset Discovery Results

No datasets were discovered during the scan.

API Feasibility Findings

Twitter API

- **Historical access:** No
- **Supports surge label construction:** Yes
- **Estimated collection time:** 0.4 hours
- **Estimated cost:** \$100.00
- **Endpoints available:** 5

Rate Limits

- free_tier: {'tweets_per_month': 1500, 'requests_per_15min': 15, 'posts_per_request': 100}
- basic_tier: {'tweets_per_month': 10000, 'requests_per_15min': 60, 'posts_per_request': 100}
- pro_tier: {'tweets_per_month': 1000000, 'requests_per_15min': 300, 'posts_per_request': 100}

Cost Tiers

Tier	Details
Free	cost_usd_monthly: 0, tweet_cap: 1500
Basic	cost_usd_monthly: 100, tweet_cap: 10000
Pro	cost_usd_monthly: 5000, tweet_cap: 1000000
Enterprise	cost_usd_monthly: 42000, tweet_cap: 50000000

Paid Fields

The following fields require paid access:

- **impression_count**: requires Basic (\$100/month)
- **full archive search**: requires Pro (\$5,000/month)
- **quote_count**: requires Basic (\$100/month)

Reddit API

- **Historical access**: Yes
- **Supports surge label construction**: Yes
- **Estimated collection time**: 0.0 hours
- **Estimated cost**: \$0.00
- **Endpoints available**: 7

Rate Limits

- `oauth_tier`: {'requests_per_minute': 100, 'posts_per_request': 100, 'daily_limit': None}
- `free_tier_note`: Reddit API is free for non-commercial use with OAuth. Commercial use requires paid access.

Cost Tiers

Tier	Details
Free (non-commercial)	<code>cost_usd_monthly</code> : 0, <code>rate_limit</code> : 100 requests/min, note: Requires OAuth app registration
Commercial	<code>cost_usd_monthly</code> : Contact Reddit, <code>rate_limit</code> : Higher limits available, note: Required for commercial data use since 2023 API changes

Paid Fields

The following fields require paid access:

- **full historical archive**: requires Commercial (contact Reddit)
- **real-time streaming**: requires Commercial (contact Reddit)

EDA Statistics

StockMarket_subreddit.csv

Dataset Structure

- **Records**: 72,620
- **Tickers**: 1597
- **Columns**: 9
- **Date range**: 1970-01-01 to 1970-01-01
- **Recommendation**: suitable

Missing Values

Column	Missing %
selftext	29.2%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	6.3	1.0	7.0	16.0	97.0
num_comments	8.1	1.0	17.0	32.0	124.0

Sentiment Statistics

- **mean:** 0.092
- **median:** 0.000
- **std:** 0.305
- **Bullish/Bearish ratio:** 2.48

Identified Risks

- Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

Surge Analysis Results

Surge definitions were evaluated across multiple threshold combinations.

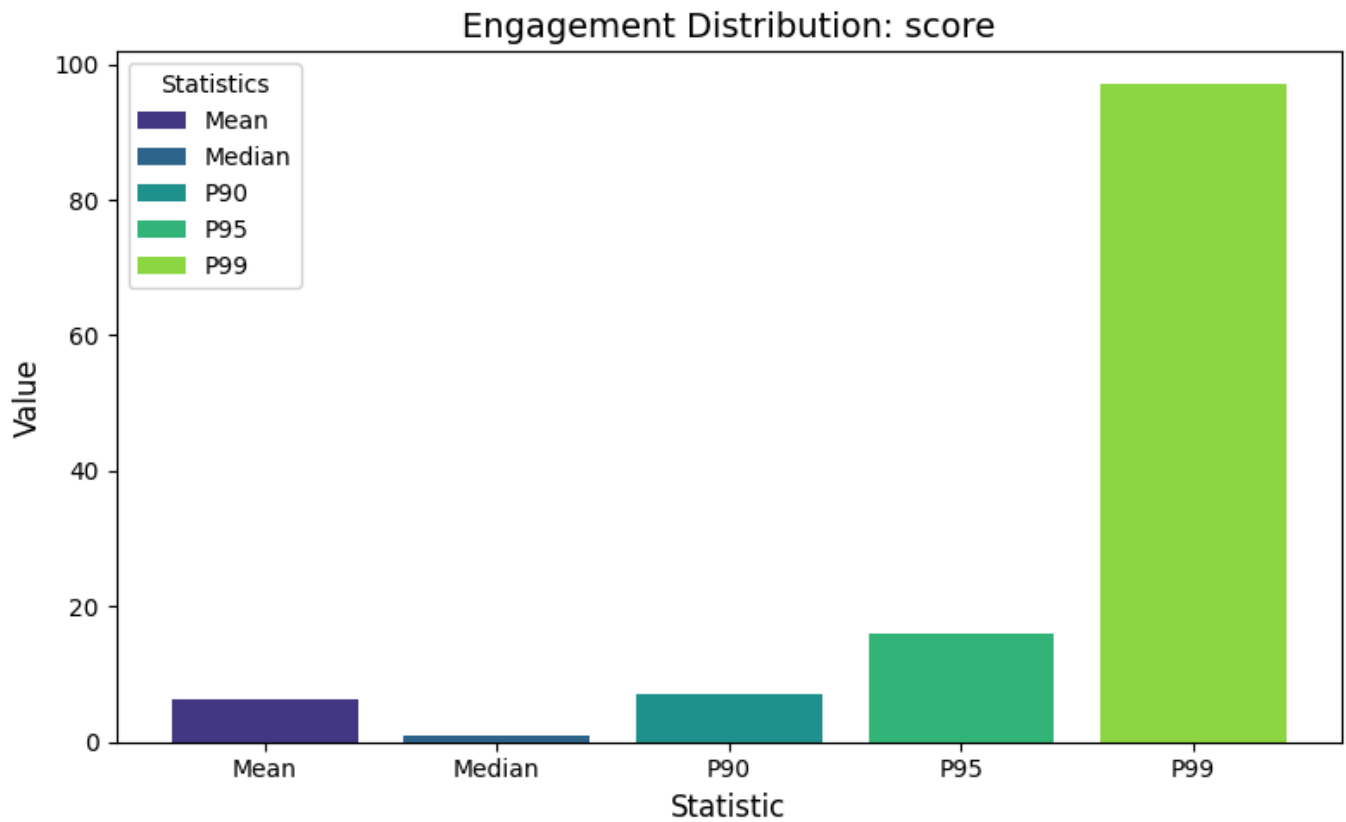
Engagement Percentile	Sentiment Std Devs	Window (hrs)	Surge Count	Total Posts	Surge %	Imbalance Ratio	Viable
0.90	0.5	24	6,171	72,620	8.50%	10.8:1	✓
0.90	1.0	24	4,411	72,620	6.07%	15.5:1	✓
0.90	1.5	24	2,483	72,620	3.42%	28.2:1	✓
0.95	0.5	24	3,521	72,620	4.85%	19.6:1	✓
0.95	1.0	24	2,388	72,620	3.29%	29.4:1	✓
0.95	1.5	24	1,371	72,620	1.89%	52.0:1	X
0.99	0.5	24	1,505	72,620	2.07%	47.3:1	✓
0.99	1.0	24	794	72,620	1.09%	90.5:1	X
0.99	1.5	24	418	72,620	0.58%	172.7:1	X

6 viable surge definition(s) found (positive class $\geq$ 2%).

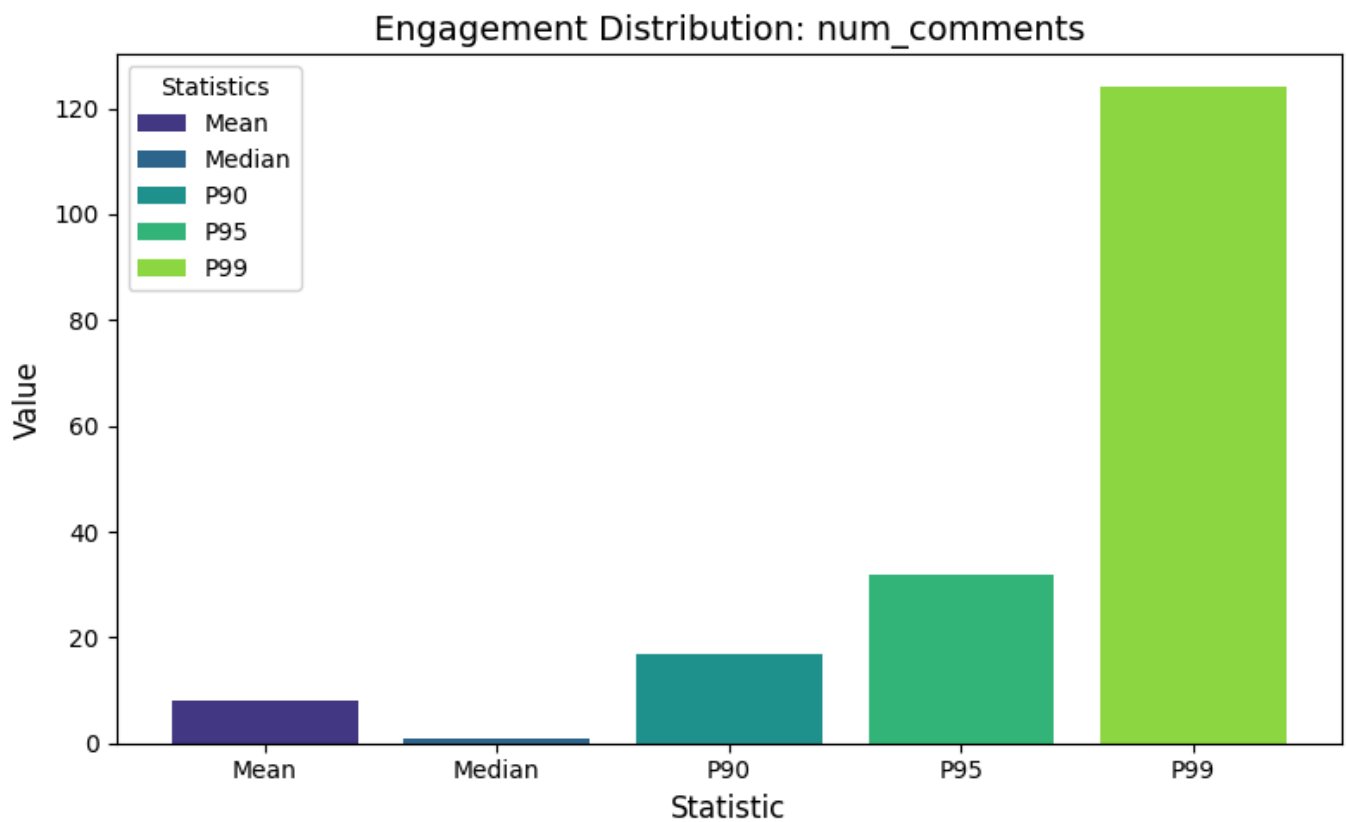
Best viable definition: engagement $\geq$ 90% percentile + sentiment shift $\geq$ 0.5 std devs $\rightarrow$ 8.50% surge rate.

Visualizations

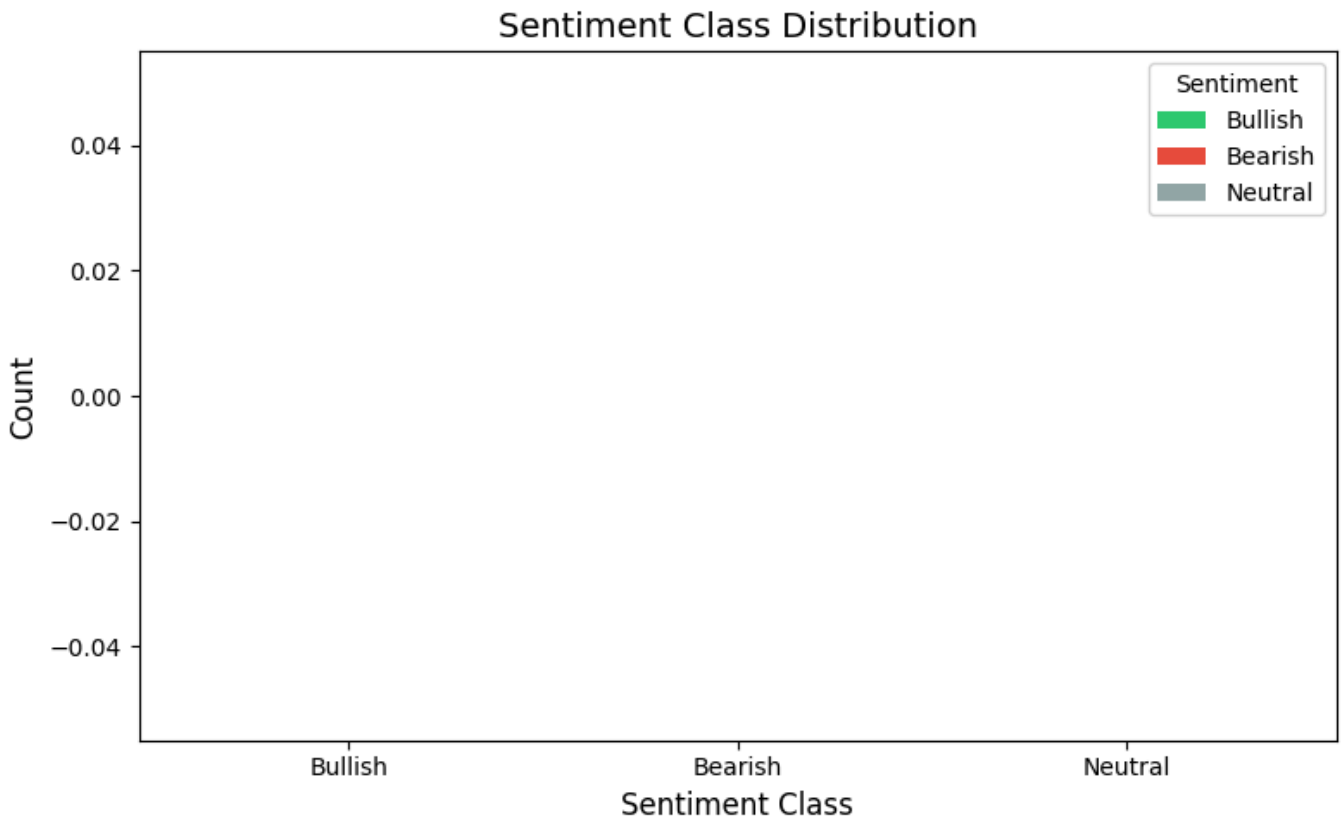
Engagement Distribution Score



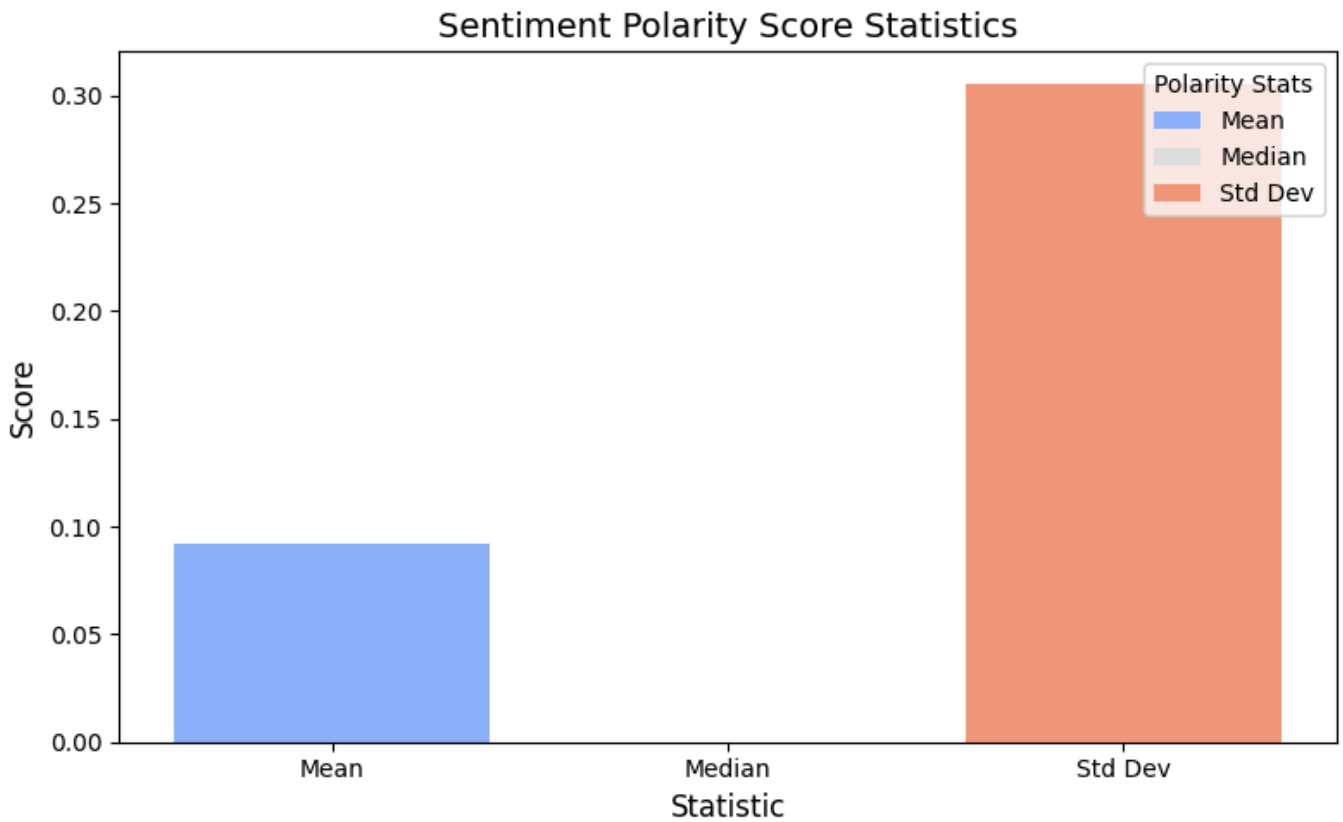
Engagement Distribution Num Comments



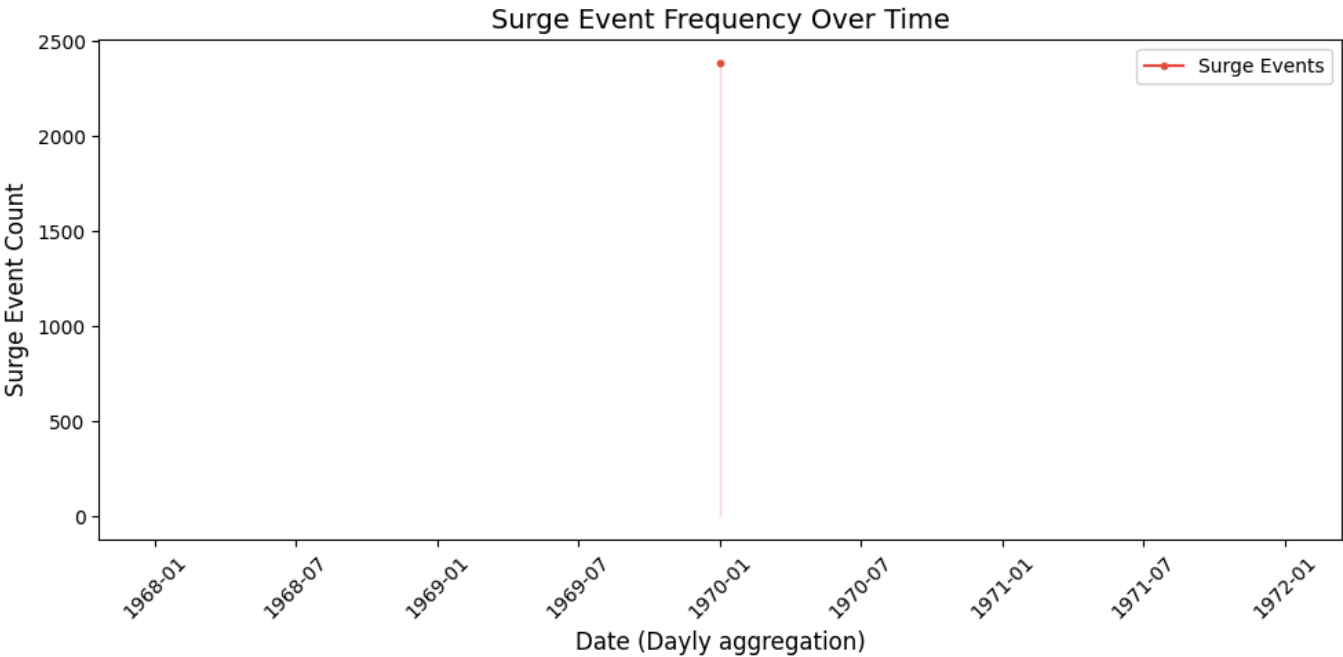
Sentiment Class Distribution



Sentiment Polarity Stats



Surge Frequency



Pipeline Performance

Stage Timings

Stage	Duration (s)
Dataset Discovery	90.93
API Feasibility	0.00
Dataset Preparation	5.95
Dataset Quality	24.55
Surge Analysis	15.26
Visualization	5.14
Total	141.84

Per-Dataset Processing Time

Dataset	Duration (s)
StockMarket_subreddit.csv	24.55

Final Recommendation

Recommended Path: API Collection (reddit API)

Recommend API collection via 'reddit API' as the best data path. Key strengths: reasonable cost, supports surge label construction, historical data access available. API collection provides fresh, customizable data tailored to the prediction task. Surge analysis confirms viable definitions exist (6/9 configurations produce $\geq 2\%$ positive class).

Ranked Options

1. **reddit API** (overall score: 1.000)

- Low cost for data collection
- Fast collection time
- Supports surge label construction
- Some fields require paid access: full historical archive, real-time streaming

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	1.00	25%	Free access
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	1.00	15%	Historical data access available
Field Availability	1.00	15%	Rich field set (10+ fields)

2. **StockMarket_subreddit.csv** (overall score: 0.930)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

3. **twitter API** (overall score: 0.770)

- Fast collection time
- Supports surge label construction
- No historical data access - requires prospective collection
- Some fields require paid access: impression_count, full archive search, quote_count

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	0.50	25%	Moderate cost (\$50-\$200)
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	0.30	15%	No historical access - prospective only
Field Availability	1.00	15%	Rich field set (10+ fields)